FermiWick

Operators, contractions, mean field, Gutzwiller projectors and matrices

Everything the package does, step by step: every output is computed by the code printed above it

Nazzareno Africani · SISSA · version 0.5.0 · November 2026

FermiWick is a Wolfram Language package for the algebra of fermion ladder operators. This document goes through all it does, in the order of a calculation: the algebra of the operators (normal order, local operators, the spin–charge basis and the recognition of spin operators); the contractions of a Gaussian state, declared by rules or taken from ready-made states, and Wick's theorem; the mean-field Hamiltonian of an operator of any order; Gutzwiller projectors; and the same operators as matrices, for exact diagonalisation. The code of each section uses only what the previous ones introduced.

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1 The algebra of the operators

1.1 Ladder operators and normal order

Fermions are $c_{i\sigma}$ and $c^\dagger_{i\sigma}$, written `c[i, s]` and `cDag[i, s]`. The site i is any expression; the spin σ is $\uparrow$, $\downarrow$ or a symbol. Products are written with `**`, in the order given, and `NormalOrder` moves every $c^\dagger$ to the left with $\{c_{i\sigma}, c^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$:

```
NormalOrder[c[1, "↑"] ** cDag[1, "↑"]]
```

$$1 - c^\dagger_{1\uparrow} c_{1\uparrow}$$

Two different symbolic sites are different sites, so their operators anticommute; a symbolic spin is split into $\uparrow$ and $\downarrow$:

```
NormalOrder[c[i, σ] ** cDag[j, σ]]
```

$$-\delta_{\downarrow,\sigma} c^\dagger_{j\downarrow} c_{i\downarrow} - \delta_{\uparrow,\sigma} c^\dagger_{j\uparrow} c_{i\uparrow}$$

```
NormalOrder[c[i, σ] ** cDag[i, σ]]
```

$$\delta_{\downarrow,\sigma}(1 - c^\dagger_{i\downarrow} c_{i\downarrow}) + \delta_{\uparrow,\sigma}(1 - c^\dagger_{i\uparrow} c_{i\uparrow})$$

Two equal fermions annihilate the product, wherever they are:

```
NormalOrder[cDag[1, "↑"] ** cDag[2, "↓"] ** cDag[1, "↑"]]
```

$$0$$

The product $a \cdot b$ (`\[CenterDot]`) is the normal-ordered product. `OperatorCommutator` and `OperatorAnticommutator` are normal-ordered too:

```
OperatorCommutator[SzOp[1], cDag[1, "↑"]]
```

$$\frac{1}{2} c^\dagger_{1\uparrow}$$

```
OperatorAnticommutator[c[1, "↑"], cDag[1, "↑"] + cDag[2, "↑"]]
```

$$1$$

1.2 Local operators, kept compact

The local operators of a site are $n_{i\sigma}$, n_i, S^z_i, $S^\pm_i$, the double occupancy $D_i = n_{i\uparrow} n_{i\downarrow}$, the doublon operators $B^\dagger_i = c^\dagger_{i\uparrow} c^\dagger_{i\downarrow}$ and B_i, and the projectors P^0_i (empty) and P^1_i (singly occupied); their definitions:

FermiWick	operator	definition
`nOp[i, s]`	$n_{i\sigma}$	$c^\dagger_{i\sigma} c_{i\sigma}$
`nTotOp[i]`, `nOp[i]`	n_i	$n_{i\uparrow} + n_{i\downarrow}$
`SzOp[i]`	S^z_i	$\frac{1}{2}\,(n_{i\uparrow} - n_{i\downarrow})$
`SpOp[i]`, `SmOp[i]`	S^+_i, S^-_i	$c^\dagger_{i\uparrow} c_{i\downarrow}$, $\;c^\dagger_{i\downarrow} c_{i\uparrow}$
`SpinOperator[i, a]`	S^a_i	$\frac{1}{2}\sum_{\sigma\sigma'} c^\dagger_{i\sigma}\,\sigma^a_{\sigma\sigma'}\,c_{i\sigma'}$, $\;a = 1,2,3$
`DOp[i]`	D_i (double occupancy)	$n_{i\uparrow} n_{i\downarrow}$
`Bdag[i]`, `B[i]`	$B^\dagger_i$, B_i (doublon)	$c^\dagger_{i\uparrow} c^\dagger_{i\downarrow}$, $\;c_{i\downarrow} c_{i\uparrow}$
`EmptyProjector[i]`	P^0_i	$(1 - n_{i\uparrow})(1 - n_{i\downarrow}) = 1 - n_i + D_i$
`SingleOccupationProjector[i]`	P^1_i	$n_i - 2D_i$
`GutzwillerProjector[i]`	P^G_i	$\sqrt{2}\,[\cos\theta\, P^1_i + \sin\theta\,(P^0_i + D_i)]$
`SdotS[i, j]`	$\mathbf{S}_i \cdot \mathbf{S}_j$	$S^z_i S^z_j + \frac{1}{2}(S^+_i S^-_j + S^-_i S^+_j)$
`SdotCrossS[i, j, k]`	$\mathbf{S}_i \cdot (\mathbf{S}_j \times \mathbf{S}_k)$	$\sum_{abc} \epsilon_{abc}\, S^a_i S^b_j S^c_k$

Products of these on one site reduce with the algebra of the fermions (e.g. $D_i n_i = 2D_i$, $(S^z_i)^2 = \frac{1}{4} n_i - \frac{1}{2} D_i$); operators on several sites and products with symbolic sites stay as products. `GutzwillerProjector` is in `FermiWick`Gutzwiller`; one-site operators on several sites are products, `DOp[i, j]` $= D_i D_j$. They are written compactly and expanded only when needed; `ToLadderBasis` expands them:

```
ToLadderBasis[SpOp[1] + SzOp[1]]
```

$$-\frac{1}{2} c^\dagger_{1\downarrow} c_{1\downarrow} + c^\dagger_{1\uparrow} c_{1\downarrow} + \frac{1}{2} c^\dagger_{1\uparrow} c_{1\uparrow}$$

`ToSpinChargeBasis` goes the other way: it writes a product with the local operators of each site. A product is separated site by site, the sites being reordered with the fermion sign only, and each site is reduced once:

```
ToSpinChargeBasis[DOp[1] ** nTotOp[1] ** SzOp[2]]
```

$$2 D_1 S^z_2$$

On one site the algebra of the local operators follows from that of the fermions: $S^+ S^- = n_\uparrow - D$, $(S^z)^2 = \frac{1}{4} n - \frac{1}{2} D$, and $\mathbf{S}^2 = \frac{3}{4}(n - 2D)$, a spin $\frac{1}{2}$ only on singly occupied sites:

```
ToSpinChargeBasis[SpOp[1] · SmOp[1]]
```

$$-D_1 + \frac{n_1}{2} + S^z_1$$

```
ToSpinChargeBasis[SzOp[1] · SzOp[1]]
```

$$\frac{n_1}{4} - \frac{D_1}{2}$$

```
ToSpinChargeBasis[SdotS[1, 1]]
```

$$\frac{3n_1}{4} - \frac{3D_1}{2}$$

The projectors are complete, and P^1 commutes with the spin:

```
ToSpinChargeBasis[EmptyProjector[1] + SingleOccupationProjector[1] + DOp[1]]
```

$$1$$

```
ToSpinChargeBasis[SingleOccupationProjector[1] · SpOp[1] ·
    SingleOccupationProjector[1] - SpOp[1]]
```

$$0$$

`GroupLocalOperators` keeps the ladder operators but groups them site by site, with the fermion sign:

```
GroupLocalOperators[cDag[1, "↑"] ** cDag[2, "↓"] ** c[2, "↓"] ** c[1, "↑"]]
```

$$c^\dagger_{1\uparrow} c_{1\uparrow} c^\dagger_{2\downarrow} c_{2\downarrow}$$

1.3 The spin–charge basis and the recognition of spin operators

A product of ladder operators on several sites is decomposed into local operators; spin flips become $S^+_1 S^-_2$:

```
ToSpinChargeBasis[cDag[1, "↑"] ** c[1, "↓"] ** cDag[2, "↓"] ** c[2, "↑"]]
```

$$S^+_1 S^-_2$$

`ToSpinChargeBasis` then recognizes the operators that act on several sites. The sum of the products of the spin components is the exchange $\mathbf{S}_1 \cdot \mathbf{S}_2$, and `"Recognize" -> False` shows its expansion:

```
ToSpinChargeBasis[Sum[SpinOperator[1, a] · SpinOperator[2, a], {a, 3}]]
```

$$\mathbf{S}_1 \cdot \mathbf{S}_2$$

```
ToSpinChargeBasis[SdotS[1, 2], "Recognize" -> False]
```

$$\frac{1}{2} S^-_1 S^+_2 + \frac{1}{2} S^+_1 S^-_2 + S^z_1 S^z_2$$

The recognition allows a remainder in the diagonal part, so that an XXZ coupling is read as a Heisenberg term plus an Ising correction:

```
ToSpinChargeBasis[J (SpinOperator[1, 1] · SpinOperator[2, 1] + SpinOperator[1, 2] ·
    SpinOperator[2, 2]) +
  Jz SzOp[1] · SzOp[2]]
```

$$J\mathbf{S}_1 \cdot \mathbf{S}_2 + (\mathrm{Jz} - J) S_1^z S_2^z$$

The commutator of two exchange terms that share a site is the scalar chirality, $[\mathbf{S}_1 \cdot \mathbf{S}_2, \mathbf{S}_2 \cdot \mathbf{S}_3] = -i\,\mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3)$; the chirality is antisymmetric in its sites and is written with them in order; and the Heisenberg triangle commutes with the total spin:

```
ToSpinChargeBasis[OperatorCommutator[SdotS[1, 2], SdotS[2, 3]]]
```

$$-i\mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3)$$

```
ToSpinChargeBasis[SdotScrossS[2, 1, 3]]
```

$$-\mathbf{S}_1 \cdot (\mathbf{S}_2 \times \mathbf{S}_3)$$

```
ToSpinChargeBasis[OperatorCommutator[SdotS[1, 2] + SdotS[2, 3] + SdotS[3, 1], SpOp[1]
    + SpOp[2] + SpOp[3]]]
```

$$0$$

These are identities of the whole Fock space, not only of the singly occupied sector.

1.4 Operators of your own

`DefineOperator` defines an operator on several sites by its expression in known operators; it stays compact, is expanded when needed and recognized in the results. A spin-conserving hopping on a bond, and its commutator with the density at its end:

```
DefineOperator[Hop, 2, (cDag[#1, "↑"] ** c[#2, "↑"] + cDag[#1, "↓"] ** c[#2, "↓"]) &,
  Subscript["T", Row[{#1, #2}]] &];
ToSpinChargeBasis[OperatorCommutator[Hop[1, 2], nTotOp[2]]]
```

$$\mathrm{T}_{12}$$

The recognition keeps the fermion signs, also with the sites out of order: the Hermitian conjugate of the hopping from 1 to 2 is the hopping from 2 to 1.

```
ToSpinChargeBasis[Dagger[Hop[1, 2]]]
```

$$\mathrm{T}_{21}$$

1.5 The Hermitian conjugate

`Dagger` reverses the products, exchanges c and $c^\dagger$ (and S^+, S^-; B, $B^\dagger$) and conjugates the coefficients. The amplitudes χ, Δ, ρ, κ are complex; the other symbols, couplings, angles and phases, are real unless declared complex with `"Complex" -> {...}`:

```
Dagger[2 I J cDag[1, SpinUp] ** c[2, SpinDown] + χ[1, 2] SpOp[1] ** Bdag[2]]
```

$$\chi_{12}^{*} B_2 S_1^{-} - 2iJ c_{2\downarrow}^{\dagger} c_{1\uparrow}$$

`PlusHC` adds the Hermitian conjugate, the "+ h.c." of a Hamiltonian, and `HermitianOperatorQ` checks it:

```
PlusHC[t E^(I \[Alpha]) cDag[1, SpinUp] ** c[2, SpinUp]]
```

$$e^{i\alpha} t c_{1\uparrow}^{\dagger} c_{2\uparrow} + e^{-i\alpha} t c_{2\uparrow}^{\dagger} c_{1\uparrow}$$

```
{HermitianOperatorQ[SdotScrossS[1, 2, 3]], HermitianOperatorQ[I SdotScrossS[1, 2, 3]]}
```

$$\{\text{True}, \text{False}\}$$

1.6 Modes with any labels

A mode can carry any labels: `c[l1, l2, ...]`. `DeclareModeLabels` says what each label is: `"Site"` (used for locality; different symbolic sites are different sites), `"Spin"` (a spin $\frac{1}{2}$, which enables the spin operators), or any other name, a generic label compared with Kronecker deltas. By default `c[i]` is a spinless fermion on site i, `c[i, s]` has a site and a spin, and more labels are a site and generic labels. Spinless fermions:

```
NormalOrder[c[1] ** cDag[2] ** c[2] ** cDag[1]]
```

$$c_2^{\dagger} c_2 + c_1^{\dagger} c_2^{\dagger} c_1 c_2$$

Orbitals: the anticommutator gives a delta of the orbitals, kept symbolic:

```
DeclareModeLabels[{"Site", "Orbital", "Spin"}];
NormalOrder[c[1, a, SpinUp] ** cDag[1, b, SpinUp]]
```

$$\delta_{a,b} - c_{1,b,\uparrow}^{\dagger} c_{1,a,\uparrow}$$

Momenta, labels with no site: the deltas stay, and multiply what they act on:

```
DeclareModeLabels[{"Momentum"}];
WickVEV[cDag[k] ** c[k] ** cDag[q] ** c[q], GaussianState[{{cDag[p1_], c[p2_]} :>
    KroneckerDelta[p1, p2] nk[p1]}]]
```

$$n_k^2(-\delta_{k,q}) + n_k\delta_{k,q} + n_k n_q$$

1.7 Coefficients and lattice phases

`OperatorCoefficients` gives the coefficient of each monomial in normal order:

```
OperatorCoefficients[2 x cDag[1, "↑"] ** c[2, "↑"] + y + c[2, "↑"] ** cDag[1, "↑"]]
```

```
<|1 -> y, cDag[1, "↑"]**c[2, "↑"] -> -1 + 2*x|>
```

`ToTeX` writes any expression in LaTeX:

```
ToTeX[ToSpinChargeBasis[OperatorCommutator[SdotS[1, 2], SdotS[2, 3]]]]
```

```
-i \mathbf S_{1}\cdot(\mathbf S_{2}\times\mathbf S_{3})
```

`DefinePeierlsHopping` and `DefineFlux` attach phases to the hoppings and fix the flux through a loop:

```
DefinePeierlsHopping[t, φ];
DefineFlux[φ, {1, 2, 3}, Φ];
Simplify[Subscript[t, 1, 2] Subscript[t, 2, 3] Subscript[t, 3, 1]]
```

$$t^3 e^{i\Phi}$$

2 Contractions and Wick's theorem

A Gaussian state is fixed by its contractions $\langle O_1 O_2 \rangle$, and Wick's theorem reduces every expectation value to them: `WickVEV` normal-orders the expression and sums over the pairings of each monomial, with the sign of the permutation. Such states, Gaussian in the fermions, are also called quasi-free. In FermiWick the state is an object, declared in one of two ways: by its symmetries, which say which contractions vanish, or by rules for the contractions.

2.1 The general state and its symmetries

A Gaussian state has one amplitude for each contraction: the hopping $\chi_{ij}^{\sigma\sigma'} = \langle c_{i\sigma}^\dagger c_{j\sigma'} \rangle$ and the pairing $\Delta_{ij}^{\sigma\sigma'} = \langle c_{i\sigma} c_{j\sigma'} \rangle$. `GaussianState[]` is the general state, with every contraction and nothing fixed, not even the filling; `ContractionTable` shows them between the modes of some sites: here $\langle c_a^\dagger c_b \rangle$ on one site, the densities on the diagonal and $\langle S_1^+ \rangle$ off it, and $\langle c_a c_b \rangle$ on two sites,

antisymmetric, so that on one site only $\langle c_{1\uparrow} c_{1\downarrow} \rangle$ is left. Hermiticity always holds: an amplitude is written with its modes in canonical order, the other order as its conjugate.

```
MatrixForm[ContractionTable[GaussianState[], {1}, "Normal"]]
```

$$\begin{pmatrix} \chi_{11}^{\uparrow\uparrow} & \chi_{11}^{\uparrow\downarrow} \\ \chi_{11}^{\uparrow\downarrow *} & \chi_{11}^{\downarrow\downarrow} \end{pmatrix}$$

```
MatrixForm[ContractionTable[GaussianState[], {1, 2}, "Anomalous"]]
```

$$\begin{pmatrix} 0 & \Delta_{11}^{\uparrow\downarrow} & \Delta_{12}^{\uparrow\uparrow} & \Delta_{12}^{\uparrow\downarrow} \\ -\Delta_{11}^{\uparrow\downarrow} & 0 & \Delta_{12}^{\downarrow\uparrow} & \Delta_{12}^{\downarrow\downarrow} \\ -\Delta_{12}^{\uparrow\uparrow} & -\Delta_{12}^{\downarrow\uparrow} & 0 & \Delta_{22}^{\uparrow\downarrow} \\ -\Delta_{12}^{\uparrow\downarrow} & -\Delta_{12}^{\downarrow\downarrow} & -\Delta_{22}^{\uparrow\downarrow} & 0 \end{pmatrix}$$

A symmetry you assume removes contractions, or the indices that told them apart, and the state is declared by its symmetries:

symmetry		what is left
`"Charge"`	particle number conserved	no pairing
`"SpinRotationZ"`	S^z conserved	the hopping keeps the spin, $\chi_{ij}^{\uparrow}$, $\chi_{ij}^{\downarrow}$; the pairing joins opposite spins, $\Delta_{ij}^{\uparrow\downarrow}$, $\Delta_{ij}^{\downarrow\uparrow}$
`"SpinRotation"`	SU(2)	one hopping χ_{ij}; the singlet $\Delta_{ij} = \langle c_{i\uparrow} c_{j\downarrow} \rangle$, symmetric in the sites
`"TimeReversal"`		the components of opposite spins conjugates of each other; with SU(2), χ and Δ real

With S^z conserved, the pairing on two sites; with SU(2) and time reversal, the hopping, real and symmetric:

```
MatrixForm[ContractionTable[GaussianState["SpinRotationZ"], {1, 2}, "Anomalous"]]
```

$$\begin{pmatrix} 0 & \Delta_{11}^{\uparrow\downarrow} & 0 & \Delta_{12}^{\uparrow\downarrow} \\ -\Delta_{11}^{\uparrow\downarrow} & 0 & \Delta_{12}^{\downarrow\uparrow} & 0 \\ 0 & -\Delta_{12}^{\downarrow\uparrow} & 0 & \Delta_{22}^{\uparrow\downarrow} \\ -\Delta_{12}^{\uparrow\downarrow} & 0 & -\Delta_{22}^{\uparrow\downarrow} & 0 \end{pmatrix}$$

```
MatrixForm[ContractionTable[GaussianState[{"SpinRotation", "TimeReversal"}], {1, 2},
    "Normal"]]
```

$$\begin{pmatrix} \chi_{11} & 0 & \chi_{12} & 0 \\ 0 & \chi_{11} & 0 & \chi_{12} \\ \chi_{12} & 0 & \chi_{22} & 0 \\ 0 & \chi_{12} & 0 & \chi_{22} \end{pmatrix}$$

The exchange in a state that conserves S^z and the particle number: the hopping of each spin, the spin flip $\chi^{\uparrow}\chi^{\downarrow *}$ and the product of the magnetizations $\frac{1}{2}(\chi^{\uparrow}_{ii} - \chi^{\downarrow}_{ii})$:

```
WickVEV[SdotS[1, 2], GaussianState[{"SpinRotationZ", "Charge"}]]
```

$$-\frac{1}{4}\chi^{\downarrow}_{12}{\chi^{\downarrow}_{12}}^{*} - \frac{1}{2}\chi^{\downarrow}_{12}{\chi^{\uparrow}_{12}}^{*} - \frac{1}{2}\chi^{\uparrow}_{12}{\chi^{\downarrow}_{12}}^{*} - \frac{1}{4}\chi^{\uparrow}_{12}{\chi^{\uparrow}_{12}}^{*} + \frac{1}{4}\chi^{\downarrow}_{11}\chi^{\downarrow}_{22} - \frac{1}{4}\chi^{\downarrow}_{22}\chi^{\uparrow}_{11} - \frac{1}{4}\chi^{\downarrow}_{11}\chi^{\uparrow}_{22} + \frac{1}{4}\chi^{\uparrow}_{11}\chi^{\uparrow}_{22}$$

The state of most calculations, invariant under spin rotations with the particle number conserved, has a short name: `SpinSingletState[]` is `GaussianState[{"SpinRotation", "Charge"}, "Filling" -> 1/2]`, the filling fixed at $\frac{1}{2}$ per spin, and `"Pairing" -> \[CapitalDelta]` drops the charge symmetry:

```
Simplify[WickVEV[SdotScrossS[1, 2, 3], GaussianState[{"SpinRotation", "Charge"},
    "Filling" -> 1/2]] -
  WickVEV[SdotScrossS[1, 2, 3], SpinSingletState[]]]
```

$$0$$

2.2 States declared by rules

`GaussianState` takes rules for the contractions, with patterns in the operators. Pairs without a rule contract to zero, and $\langle c\, c^{\dagger}\rangle$ follows from $\langle c^{\dagger} c\rangle$ by the anticommutator. Two sites with hopping g and half filling:

```
dimer = GaussianState[{{cDag[i_, s_], c[j_, t_]} :> KroneckerDelta[s, t] If[i === j,
    1/2, g]}];
WickVEV[nOp[1, "↑"] ** nOp[2, "↑"], dimer]
```

$$\frac{1}{4} - g^2$$

```
{Contraction[dimer, cDag[1, "↑"], c[2, "↑"]], Contraction[dimer, c[1, "↑"], cDag[2,
    "↑"]]}
```

$$\{g, -g\}$$

The rules can have any spin structure: here a hopping h that flips the spin, besides the hopping g that keeps it; it enters the correlation of S^z with the opposite sign:

```
flip = GaussianState[{{cDag[i_, s_], c[j_, t_]} :> If[i === j, KroneckerDelta[s,
    t]/2, If[s === t, g, h]]}];
WickVEV[SzOp[1] ** SzOp[2], flip]
```

$$\frac{h^2}{2} - \frac{g^2}{2}$$

2.3 States invariant under spin rotations

In SpinSingletState, with pairing, hopping and pairing enter the exchange with the same weight, as the SU(2) gauge structure of spins written with fermions requires [1]:

```
state = SpinSingletState["Pairing" -> Δ]
```

```
GaussianState["spin singlet with pairing"]
```

```
WickVEV[SdotS[1, 2], state]
```

$$-\frac{3}{2}\Delta_{12}\Delta_{12}^{*}-\frac{3}{2}\chi_{12}\chi_{12}^{*}$$

The chirality is a closed loop of three contractions, three hoppings or one hopping and two pairings; its imaginary part is the flux of the amplitudes through the triangle:

```
WickVEV[SdotScrossS[1, 2, 3], state]
```

$$\frac{3}{2}i\Delta_{13}\Delta_{12}^{*}\chi_{23}^{*}-\frac{3}{2}i\Delta_{13}\chi_{12}\Delta_{23}^{*}+\frac{3}{2}i\Delta_{23}\Delta_{13}^{*}\chi_{12}^{*}-\frac{3}{2}i\Delta_{23}\Delta_{12}^{*}\chi_{13}^{*}+\frac{3}{2}i\Delta_{12}\chi_{13}\Delta_{23}^{*}-\frac{3}{2}i\Delta_{12}\chi_{23}\Delta_{13}^{*}+\frac{3}{2}i\chi_{13}\chi_{12}^{*}\chi_{23}^{*}-\frac{3}{2}i\chi_{12}\chi_{23}\chi_{13}^{*}$$

Filling and on-site pairing are options:

```
WickVEV[DOp[1], SpinSingletState["Filling" -> ν, "Pairing" -> Δ, "OnSitePairing" -> η
    ]]
```

$$\eta\eta^{*}+\nu^{2}$$

2.4 Magnetic and general states

A state need not be invariant. Local magnetizations m_i, $\langle n_{i\sigma}\rangle = \frac{1}{2} \pm m_i$, with no contractions between sites, give the classical exchange:

```
sgn["↑"] = 1; sgn["↓"] = -1;
neel = GaussianState[{{cDag[i_, s_], c[i_, t_]} :> KroneckerDelta[s, t] (1/2 + sgn[s]
    m[i])}]
```

```
GaussianState["user-defined"]
```

```
WickVEV[SdotS[1, 2], neel]
```

$$m_1 m_2$$

The same state follows from the symmetry: rotations about z, the magnetization m_i on each site, and only the local contractions:

```
WickVEV[SdotS[1, 2], StateSelect[GaussianState["SpinRotationZ", "Magnetization" ->
    (m[#] &)], LocalPairQ]]
```

$$m_1 m_2$$

The general state has every contraction: a transverse magnetization on one site, $\langle S_1^+ \rangle = \chi_{11}^{\uparrow\downarrow}$, and the pairing between two sites in the correlation of two pairs:

```
WickVEV[SpOp[1] + SzOp[1], GaussianState[]]
```

$$-\frac{1}{2}\chi_{11}^{\downarrow\downarrow} + \chi_{11}^{\uparrow\downarrow} + \frac{1}{2}\chi_{11}^{\uparrow\uparrow}$$

```
WickVEV[Bdag[1] ** B[2], GaussianState[]]
```

$$\Delta_{22}^{\uparrow\downarrow}\Delta_{11}^{\uparrow\downarrow *} - \chi_{12}^{\downarrow\uparrow}\chi_{12}^{\uparrow\downarrow} + \chi_{12}^{\downarrow\downarrow}\chi_{12}^{\uparrow\uparrow}$$

2.5 Selecting and truncating contractions

Symmetries describe states; filters select terms. `StateSelect` keeps the contractions that satisfy a criterion: the result is a physical state when the criterion is a symmetry or a decoupling of the sites, and otherwise a way to isolate a class of terms. Without the anomalous ones the pairing state is a pure-hopping state; with only the local ones the sites are uncorrelated; with only the non-local ones the Hartree term is gone:

```
WickVEV[SdotS[1, 2], StateSelect[state, ! AnomalousPairQ[##] &]]
```

$$-\frac{3}{2}\chi_{12}\chi_{12}^{*}$$

```
WickVEV[nTotOp[1] ** nTotOp[2], StateSelect[SpinSingletState[], LocalPairQ]]
```

$$1$$

```
WickVEV[nTotOp[1] ** nTotOp[2], StateSelect[SpinSingletState[], ! LocalPairQ[##] &]]
```

$$-2\chi_{12}\chi_{12}^{*}$$

`"MaxOrder" -> m` keeps the terms with at most m contractions between different sites, and `VerifyNormalOrder` checks Wick's theorem with and without normal ordering:

```
WickVEV[SdotS[1, 2] ** SdotS[2, 3], "MaxOrder" -> 2]
```

$$-\frac{3}{16}\chi_{13}\chi_{13}^{*}$$

```
VerifyNormalOrder[SdotS[1, 2] ** SdotS[2, 3], state]
```

$$0$$

2.6 Wick's theorem written out

`WickPairings` lists the pairings of a product that do not vanish, with their signs and values; `WickTeX` writes the whole expansion in LaTeX, with the contractions drawn by the package `simpler-wick` (two LaTeX runs). The pairings of a hopping back and forth, at half filling:

```
WickPairings[cDag[1, SpinUp] ** c[2, SpinUp] ** cDag[2, SpinUp] ** c[1, SpinUp]]
```

```
{<|"Sign" -> 1, "Pairs" -> {{1, 2}, {3, 4}}, "Value" -> Conjugate[χ[1, 2]]*χ[1,
    2]|>,
 <|"Sign" -> 1, "Pairs" -> {{1, 4}, {2, 3}}, "Value" -> 1/4|>}
```

Hopping and pairing in a product of four operators: the string, and the string typeset:

```
WickTeX[cDag[1, SpinUp] ** cDag[2, SpinDown] ** c[3, SpinDown] ** c[4, SpinUp], state]
```

```
\langle c^\dagger_{1\uparrow} c^\dagger_{2\downarrow} c_{3\downarrow}
    c_{4\uparrow}\rangle = \wick{\c1 c^\dagger_{1\uparrow} \c1
    c^\dagger_{2\downarrow} \c1 c_{3\downarrow} \c1 c_{4\uparrow}} + \wick{\c2
    c^\dagger_{1\uparrow} \c1 c^\dagger_{2\downarrow} \c1 c_{3\downarrow} \c2
    c_{4\uparrow}} = \Delta_{34} \Delta_{12}^*+\chi_{14} \chi_{23}
```

$$\langle c^\dagger_{1\uparrow} c^\dagger_{2\downarrow} c_{3\downarrow} c_{4\uparrow}\rangle = \overbrace{c^\dagger_{1\uparrow} c^\dagger_{2\downarrow}}\, \overbrace{c_{3\downarrow} c_{4\uparrow}} + \overbrace{c^\dagger_{1\uparrow} \overbrace{c^\dagger_{2\downarrow} c_{3\downarrow}} c_{4\uparrow}} = \Delta_{34} \Delta_{12}^* + \chi_{14} \chi_{23}$$

Only the non-local contractions of a loop around a triangle; overlapping contractions are drawn at different heights:

```
WickTeX[cDag[1, SpinUp] ** c[2, SpinUp] ** cDag[2, SpinUp] ** c[3, SpinUp] ** cDag[3,
    SpinUp] ** c[1, SpinUp],
  StateSelect[state, ! LocalPairQ[##] &]]
```

```
\langle c^\dagger_{1\uparrow} c_{2\uparrow} c^\dagger_{2\uparrow} c_{3\uparrow}
    c^\dagger_{3\uparrow} c_{1\uparrow}\rangle = \wick{\c1 c^\dagger_{1\uparrow}
    \c1 c_{2\uparrow} \c1 c^\dagger_{2\uparrow} \c1 c_{3\uparrow} \c1
    c^\dagger_{3\uparrow} \c1 c_{1\uparrow}} - \wick{\c1 c^\dagger_{1\uparrow} \c2
    c_{2\uparrow} \c3 c^\dagger_{2\uparrow} \c1 c_{3\uparrow} \c2
    c^\dagger_{3\uparrow} \c3 c_{1\uparrow}} = \chi_{12} \chi_{23} \chi_{13}^* +
    \chi_{13} \chi_{12}^* \chi_{23}^* = \chi_{13} \chi_{12}^*
    \chi_{23}^*+\chi_{12} \chi_{23} \chi_{13}^*
```

$$\langle c^\dagger_{1\uparrow} c_{2\uparrow} c^\dagger_{2\uparrow} c_{3\uparrow} c^\dagger_{3\uparrow} c_{1\uparrow}\rangle = \overbrace{c^\dagger_{1\uparrow} c_{2\uparrow}}\, \overbrace{c^\dagger_{2\uparrow} c_{3\uparrow}}\, \overbrace{c^\dagger_{3\uparrow} c_{1\uparrow}} - \overbrace{c^\dagger_{1\uparrow} c_{2\uparrow} c^\dagger_{2\uparrow} c_{3\uparrow} c^\dagger_{3\uparrow} c_{1\uparrow}} = \chi_{12}\chi_{23}\chi_{13}^* + \chi_{13}\chi_{12}^*\chi_{23}^* = \chi_{13}\chi_{12}^*\chi_{23}^* + \chi_{12}\chi_{23}\chi_{13}^*$$

3 Mean field

3.1 The mean-field Hamiltonian of any operator

The Hartree–Fock–Bogoliubov Hamiltonian is the quadratic operator whose expectation value has the same first variation as $\mathcal{E} = \langle H \rangle$ among Gaussian states. For a product of $2n$ fermions it is obtained by taking two operators out in every possible way, in their order and with the sign of the permutation that brings them to the front, and replacing the rest by its expectation value; the constant $-(n-1)\langle O_1 \cdots O_{2n} \rangle$ gives $\langle H_{\mathrm{MF}} \rangle = \mathcal{E}$. `MeanField` does this for any order and any state. The Hubbard interaction gives Hartree and BCS terms in the singlet state, and a spin-dependent Hartree field in the magnetic one:

```
MeanField[U DOp[1], SpinSingletState["Filling" -> ν, "Pairing" -> Δ, "OnSitePairing"
    -> η]]
```

$$\nu U c^\dagger_{1\downarrow} c_{1\downarrow} + \nu U c^\dagger_{1\uparrow} c_{1\uparrow} - U\eta^* c_{1\downarrow} c_{1\uparrow} + \eta U c^\dagger_{1\downarrow} c^\dagger_{1\uparrow} - \eta U \eta^* + \nu^2(-U)$$

```
MeanField[U DOp[1], neel]
```

$$m_1 U c^\dagger_{1\downarrow} c_{1\downarrow} - m_1 U c^\dagger_{1\uparrow} c_{1\uparrow} + \frac{1}{2} U c^\dagger_{1\downarrow} c_{1\downarrow} + \frac{1}{2} U c^\dagger_{1\uparrow} c_{1\uparrow} + m_1^2 U - \frac{U}{4}$$

A density–density interaction has Hartree and Fock terms; a quadratic operator is its own mean field:

```
MeanField[nTotOp[1] ** nTotOp[2]]
```

$$-\chi^*_{12} c^\dagger_{1\downarrow} c_{2\downarrow} - \chi^*_{12} c^\dagger_{1\uparrow} c_{2\uparrow} - \chi_{12} c^\dagger_{2\downarrow} c_{1\downarrow} - \chi_{12} c^\dagger_{2\uparrow} c_{1\uparrow} + c^\dagger_{1\downarrow} c_{1\downarrow} + c^\dagger_{2\downarrow} c_{2\downarrow} + c^\dagger_{1\uparrow} c_{1\uparrow} + c^\dagger_{2\uparrow} c_{2\uparrow} + 2\chi_{12}\chi^*_{12} - 1$$

```
MeanField[cDag[1, "↑"] ** c[2, "↑"] + cDag[2, "↑"] ** c[1, "↑"]]
```

$$c^\dagger_{1\uparrow} c_{2\uparrow} + c^\dagger_{2\uparrow} c_{1\uparrow}$$

3.2 The fields of a Hamiltonian

`OperatorCoefficients` reads the fields. For the exchange $J\,\mathbf{S}_1 \cdot \mathbf{S}_2$: the hopping field $t_{12} = -\frac{3}{4} J \chi^*_{12}$, the pairing field $D_{12} = \frac{3}{4} J \Delta_{12}$ and the constant:

```
mf = MeanField[J SdotS[1, 2], state];
fields = OperatorCoefficients[mf];
{fields[cDag[1, "↑"] ** c[2, "↑"]], fields[cDag[1, "↑"] ** cDag[2, "↓"]], fields[1]}
```

$$\left\{ -\frac{3}{4} J \chi^*_{12}, \frac{3}{4} J \Delta_{12}, \frac{3}{2} J \Delta_{12} \Delta^*_{12} + \frac{3}{2} J \chi_{12} \chi^*_{12} \right\}$$

The chirality has six fermions, so its fields are quadratic in the amplitudes: each bond of the

triangle receives a hopping from the path through the third site, and a pairing from one hopping and one pairing on the other two sides:

```
mfc = OperatorCoefficients[MeanField[-Jc SdotScrossS[1, 2, 3], state]];
{mfc[cDag[1, "↑"] ** c[2, "↑"]], mfc[cDag[1, "↑"] ** cDag[2, "↓"]]}
```

$$\left\{\frac{3}{4}iJ_\chi\Delta_{13}\Delta_{23}^* + \frac{3}{4}iJ_\chi\chi_{23}\chi_{13}^*, \frac{3}{4}iJ_\chi\Delta_{13}\chi_{23}^* - \frac{3}{4}iJ_\chi\Delta_{23}\chi_{13}^*\right\}$$

3.3 Hermiticity and the variational property

The mean-field Hamiltonian of a Hermitian operator is Hermitian, and has the expectation value of the operator, identically:

```
HermitianOperatorQ[MeanField[-Jc SdotScrossS[1, 2, 3], state]]
```

True

```
Simplify[WickVEV[MeanField[SdotScrossS[1, 2, 3], state], state] -
    WickVEV[SdotScrossS[1, 2, 3], state]]
```

0

4 Gutzwiller projectors

`FermiWick`Gutzwiller`` adds the half-filling Gutzwiller operator $P_i^G = \sqrt{2}\,[\cos\theta\, P_i^1 + \sin\theta\,(P_i^0 + D_i)]$, with $\theta = 0$ the projection on single occupation, and `GutzwillerVEV`, the expectation value to first order in the contractions between different sites. In the spin–charge basis:

```
ToSpinChargeBasis[GutzwillerProjector[1]]
```

$$D_1\left(2\sqrt{2}\sin(\theta) - 2\sqrt{2}\cos(\theta)\right) + \sqrt{2}\sin(\theta) + n_1\left(\sqrt{2}\cos(\theta) - \sqrt{2}\sin(\theta)\right)$$

At half filling P^G is normalized for every θ, and the double occupancy is $d = \frac{1}{2}\sin^2\theta$:

```
GutzwillerVEV[GutzwillerProjector[1] · GutzwillerProjector[1]]
```

1

```
GutzwillerVEV[LocalSandwich[GutzwillerProjector, {1}, DOp[1]]]
```

$$\frac{\sin^2(\theta)}{2}$$

In a state with real hopping (time reversal), the hopping between two projected sites is reduced by

```
real = GaussianState[{"SpinRotation", "Charge", "TimeReversal"}, "Filling" -> 1/2];
hop = Sum[cDag[1, s] ** c[2, s] + cDag[2, s] ** c[1, s], {s, $Spins}];
Simplify[GutzwillerVEV[LocalSandwich[GutzwillerProjector, {1, 2}, hop],
    real]/WickVEV[hop, real]]
```

$$\sin^2(2\theta)$$

which is the factor $q = 8d(1-2d)$ of Brinkman and Rice [2]:

```
Simplify[Sin[2 θ]^2 - 8 d (1 - 2 d) /. d -> Sin[θ]^2/2]
```

$$0$$

Minimizing $8d(1-2d)\,\varepsilon_0 + Ud$, with $\varepsilon_0 < 0$ the kinetic energy of the free electrons per bond, gives

```
d /. First[Solve[D[eps0 8 d (1 - 2 d) + U d, d] == 0, d]]
```

$$\frac{8\varepsilon_0 + U}{32\varepsilon_0}$$

so that d vanishes at $U_c = 8|\varepsilon_0|$, the Brinkman–Rice transition. The exchange, instead, is enhanced: for $\theta = 0$ it is four times its value in the free state, the factor $g_s = 4$ of Zhang, Gros, Rice and Shiba at half filling [3]:

```
Simplify[WickVEV[LocalSandwich[GutzwillerProjector, {1, 2}, SdotS[1, 2]]]]
```

$$-6\chi_{12}\cos^4(\theta)\chi_{12}^*$$

5 Matrices and exact diagonalisation

`FermiWick`Fock`` writes the operators of the previous sections as sparse matrices on the Fock space of a few sites, so that a cluster can be solved exactly and a symbolic result checked. It is loaded, in a cell of its own, by

```
Needs["FermiWick`Fock`"]
```

A space is given by its sites (and orbitals, or no spin); its modes are the ladder operators, numbered site by site with the spin fastest, and a basis state is the string of their occupations, the first mode being the most significant bit:

```
space = FockSpace[{1, 2}];
space["Modes"]
```

```
{c[1, "↑"], c[1, "↓"], c[2, "↑"], c[2, "↓"]}
```

A sector keeps the states with a given particle number and S^z, and every operator is built directly in it, without the whole space:

```
sector = FockSector[space, "N" -> 2, "Sz" -> 0];
OccupationBasis[sector]
```

```
{{0, 0, 1, 1}, {0, 1, 1, 0}, {1, 0, 0, 1}, {1, 1, 0, 0}}
```

`FockMatrix` reads any expression of the package: ladder, local and defined operators, products in the order written, symbolic coefficients. The Hubbard dimer, written as on paper with the interaction in the form that is symmetric about half filling, $\frac{U}{2}\sum_i (n_i - 1)^2$, has in this sector the spectrum

```
H = -t Sum[PlusHC[cDag[1, s] ** c[2, s]], {s, $Spins}] + U/2 Sum[(nTotOp[i] - 1)^2,
    {i, 2}];
Eigenvalues[Normal[FockMatrix[H, sector]]]
```

$$\left\{0, U, \frac{1}{2}\left(U - \sqrt{16t^2 + U^2}\right), \frac{1}{2}\left(\sqrt{16t^2 + U^2} + U\right)\right\}$$

The eigenvectors are exact as well, and `StateToKet` writes a state with the occupations of the sites. The ground state, the lowest of the four energies for all $t, U > 0$ (the numbers only rank them), is a singlet whose weight on the doubly occupied sites falls as U grows:

```
{vals, vecs} = Eigensystem[Normal[FockMatrix[H, sector]]];
ground = Normalize[vecs[[First[Ordering[N[vals /. {t -> 1, U -> 1}], 1]]]]];
FullSimplify[StateToKet[sector, ground], t > 0 && U > 0]
```

$$\frac{4t(|0,\uparrow\downarrow\rangle + |\uparrow\downarrow,0\rangle) - \left(\sqrt{16t^2 + U^2} + U\right)(|\downarrow,\uparrow\rangle - |\uparrow,\downarrow\rangle)}{2\sqrt{U\left(\sqrt{16t^2 + U^2} + U\right) + 16t^2}}$$

At a finite temperature everything follows from the grand-canonical partition function $Z = \operatorname{Tr} e^{-(H-\mu N)/T}$, the trace over the whole Fock space, here sixteen states. `PartitionFunction` gives it exactly, as a sum of exponentials over the levels of every sector:

```
Z = PartitionFunction[H, space, T, μ]
```

$$e^{\frac{2\mu+\frac{1}{2}\left(-\sqrt{16t^2+U^2}-U\right)}{T}} + e^{\frac{2\mu+\frac{1}{2}\left(\sqrt{16t^2+U^2}-U\right)}{T}} + 2e^{\frac{\mu+\frac{1}{2}(2t-U)}{T}} + 2e^{\frac{\mu-t-\frac{U}{2}}{T}} + 2e^{\frac{3\mu+\frac{1}{2}(2t-U)}{T}} + 2e^{\frac{3\mu-t-\frac{U}{2}}{T}} + 3e^{\frac{2\mu}{T}} + e^{\frac{2\mu-U}{T}} + e^{\frac{4\mu-U}{T}} + e^{-\frac{U}{T}}$$

Its logarithm is the grand potential $\Omega = -T \log Z$, and the physical quantities are its derivatives. The particle number $-\partial\Omega/\partial\mu$ is two at $\mu = 0$, half filling, by the symmetry of the interaction; its derivative is the compressibility:

```
Ω = -T Log[Z];
FullSimplify[{-D[Ω, μ], -D[Ω, {μ, 2}]} /. μ -> 0, T > 0]
```

$$\left\{2, \frac{4\left(\cosh\left(\frac{t}{T}\right) + e^{-\frac{U}{2T}}\right)}{T\left(\cosh\left(\frac{\sqrt{16t^2+U^2}}{2T}\right) + 4\cosh\left(\frac{t}{T}\right) + 3\cosh\left(\frac{U}{2T}\right)\right)}\right\}$$

At half filling the free energy is

```
F = FullSimplify[-T Log[PartitionFunction[H, space, T]], T > 0]
```

$$-T \log\left(2e^{-\frac{U}{2T}}\left(\cosh\left(\frac{\sqrt{16t^2+U^2}}{2T}\right) + 4\cosh\left(\frac{t}{T}\right) + 3\cosh\left(\frac{U}{2T}\right)\right)\right)$$

and the entropy $S = -\partial F/\partial T$ goes from zero, the ground state being a singlet, to $\log 16$, the sixteen states equally likely (at $U = 4t$ for the first limit):

```
S = -D[F, T];
{Limit[S /. U -> 4 t, T -> 0, Assumptions -> t > 0], Limit[S, T -> Infinity]}
```

$$\{0, \log(16)\}$$

The single-particle Green's function is a sum over the same levels, the Lehmann representation, $G_{ab}(z) = \sum_{nm}(w_n + w_m)\langle n|c_a|m\rangle\langle m|c_b^\dagger|n\rangle/(z - E_m + E_n)$, with w_n the weight of a state. `LehmannGreen` takes the Hamiltonian, the space and the modes, and returns its poles and residues; `GreenMatrix` evaluates it. At zero temperature, in closed form:

```
g = LehmannGreen[H, space, {c[1, "↑"]}];
FullSimplify[GreenMatrix[g, z][[1, 1]], t > 0 && U > 0]
```

$$-\frac{4z\left(36t^2 + U^2 - 4z^2\right)}{-8z^2\left(20t^2 + U^2\right) + \left(12t^2 + U^2\right)^2 + 16z^4}$$

Its four poles, at $\pm\left(\frac{1}{2}\sqrt{16t^2 + U^2} \pm t\right)$, are the energies of adding and of removing an electron. At

a temperature T every state has the weight $e^{-E_n/T}/Z$, and the transitions between excited states add four poles:

```
gT = LehmannGreen[H, space, {c[1, "↑"]}, "Temperature" -> T];
Complement[gT["Poles"], g["Poles"]]
```

$$\left\{-t-\frac{U}{2}, t-\frac{U}{2}, t+\frac{U}{2}, \frac{1}{2}(U-2t)\right\}$$

Each of the new poles has the same weight, which vanishes as $T \to 0$:

```
weights = AssociationThread[gT["Poles"], gT["Residues"][[All, 1, 1]]];
FullSimplify[weights[t + U/2], T > 0]
```

$$\frac{3\left(\cosh\left(\frac{t}{T}\right)+\cosh\left(\frac{U}{2T}\right)\right)}{4\left(\cosh\left(\frac{\sqrt{16t^2+U^2}}{2T}\right)+4\cosh\left(\frac{t}{T}\right)+3\cosh\left(\frac{U}{2T}\right)\right)}$$

Ground states come from `LowestStates` (dense, or Arnoldi on large sectors) and expectation values from `FockExpectation`. On a ring of eight sites at half filling with $U = 4t$ (a sector of dimension 4900), the energy per site, the double occupancy and the correlation of neighbouring spins are

```
nb[i_] := Mod[i - 1, 8] + 1;
hop8 = -Sum[cDag[i, s] ** c[nb[i + 1], s] + cDag[nb[i + 1], s] ** c[i, s], {i, 8},
    {s, $Spins}];
half = FockSector[FockSpace[Range[8]], "N" -> 8, "Sz" -> 0];
{e, v} = LowestStates[FockMatrix[hop8 + 4 Sum[DOp[i], {i, 8}], half], 1];
N[Round[10^4 {First[e]/8, FockExpectation[DOp[1], First[v], half],
    FockExpectation[SdotS[1, 2], First[v], half]}]/10^4]
```

```
{-0.5754, 0.0949, -0.34}
```

With numbers, `LehmannGreen` and `GrandPotential` work on larger clusters: a ring of four sites at $U = 4t$ and $T = t/2$, a Fock space of 256 states: its grand potential, the number of poles of its local Green's function, and the imaginary part of the latter at the first Matsubara frequency, $z = i\pi T$:

```
ring4 = -Sum[PlusHC[cDag[i, s] ** c[Mod[i, 4] + 1, s]], {i, 4}, {s, $Spins}] + 2
    Sum[(nTotOp[i] - 1)^2, {i, 4}];
g4 = LehmannGreen[ring4, FockSpace[Range[4]], {c[1, "↑"]}, "Temperature" -> 1/2];
{N[Round[10^4 g4["GrandPotential"]]/10^4], Length[g4["Poles"]], N[Round[10^4
    Im[GreenMatrix[g4, I Pi/2][[1, 1]]]]/10^4]}
```

```
{-2.8025, 165, -0.2831}
```

Wick's theorem is exact in a Gaussian state, and here it can be checked: the contractions of the free ground state of the ring with six electrons (a closed shell, so that the ground state is a single Slater

determinant), computed from the matrices, define a GaussianState, and WickVEV of an interacting operator in that state is its exact expectation value:

```
six = FockSector[FockSpace[Range[8]], "N" -> 6, "Sz" -> 0];
v0 = First[Last[LowestStates[FockMatrix[hop8, six], 1]]];
pairs = Flatten[Table[{cDag[i, s], c[j, s]}, {i, 8}, {j, 8}, {s, $Spins}], 2];
exact = GaussianState[Thread[pairs -> (FockExpectation[#1 ** #2, v0, six] & @@@
    pairs)]];
op = SdotS[1, 2] + DOp[1];
Chop[WickVEV[op, exact] - FockExpectation[op, v0, six]]
```

```
0
```

6 Checks

The test suite (`Tests/RunTests.wls`) checks the algebra, the mean-field formulas of the exchange, of the chirality and of the Hubbard interaction, the states, and the identities of this showcase. It checks the symbolic engine against the matrices: the matrix of an expression equals that of its normal order and of its spin–charge form, and WickVEV and MeanField give the exact values in a random Gibbs state of a quadratic Hamiltonian with pairing, on three sites with spin and on two sites with two orbitals. It checks the matrices themselves against the Jordan–Wigner construction, and exact diagonalisation on the Hubbard dimer, a magnetic plaquette and Hubbard rings: the residuals are below 10^{-10}.

References

[1] I. Affleck, Z. Zou, T. Hsu, and P. W. Anderson, Phys. Rev. B **38**, 745 (1988).

[2] W. F. Brinkman and T. M. Rice, Phys. Rev. B **2**, 4302 (1970).

[3] F. C. Zhang, C. Gros, T. M. Rice, and H. Shiba, Supercond. Sci. Technol. **1**, 36 (1988).